⚙ CLAMPFORCE v1.0

User Manual

Bolted Flange Connection Design Tool · Pressure Vessels & Piping

Version	Standards	Platform
v1.0	VDI 2230 · EN 1591	Web (HTML) — no installation required

1. General Overview

ClampForce v1.0 is an engineering calculation tool for analysing bolted flange connections in pressure vessels and piping systems in accordance with VDI 2230 and EN 1591. The tool runs entirely in a web browser — no installation, internet connection, or third-party software is required.

> ℹ **The tool consists of a single .html file. Double-click it to launch on any computer,** with or without an internet connection.

What the Tool Does

- Checks the stress capacity of bolts in a flanged joint
- Verifies that the joint will not separate under pressure (separation check)
- Optionally calculates gasket seating stress for leak tightness
- Optionally accounts for additional loads from differential thermal expansion
- In DESIGN mode, automatically determines the minimum bolt count for given operating conditions

When to Use It

- Verifying the adequacy of an existing flanged joint (CHECK mode)
- Determining bolt count and type for a new flanged joint (DESIGN mode)
- Evaluating flange behaviour under varying temperature conditions
- Calculating gasket material requirements and sealing margin

2. Interface Layout

The tool is divided into two main panels: the input panel on the left and the results panel on the right.

LEFT PANEL — Parameter Input	RIGHT PANEL — Results
• Pressure & Geometry • Bolt Properties • Safety Factors • Stiffness Model (toggle) • Thermal Loading (toggle) • Gasket Seating (toggle)	• Overall Accept / Reject status • 6 key metric cards • Stress check (progress bar) • Separation check • Gasket seating check (if active) • Bolt spacing check (DESIGN mode)

Mode Selection: CHECK and DESIGN

Two buttons in the top-right corner of the screen select the operating mode.

Mode	When to Use	Additional Field
CHECK	Verifying an existing flanged joint. Bolt count is entered by the user.	Bolt Count (n)
DESIGN	New joint design. The tool calculates the minimum bolt count for the given conditions.	BCD — Bolt Circle Diameter (mm)

3. Input Parameters

3.1 Pressure & Geometry

This section defines the fundamental physical conditions of the system.

Parameter	Default	Description
Internal Pressure (MPa)	0.25	Design pressure inside the flange. Enter gauge (manometric) pressure.
Bolt Count (n)	8	Visible in CHECK mode only. Total number of bolts on the flange.
BCD — Bolt Circle (mm)	240	Visible in DESIGN mode only. Bolt circle diameter; diameter of the circle passing through bolt centrelines.
Gasket Inner Diameter (mm)	180	Inner edge diameter of the gasket — the inner boundary of the sealing surface.
Gasket Outer Diameter (mm)	220	Outer edge diameter of the gasket. The average of inner and outer diameters gives the effective gasket diameter.

ℹ **The tool calculates the effective pressure area using: $A_{eff} = \pi/4 \times d_{eff}^2$, where $d_{eff} = (d_{in} + d_{out}) / 2$.**
The inner diameter must always be less than the outer diameter — an error message will appear otherwise.

3.2 Bolt Properties

Bolt geometry and material data are entered in this section.

Parameter	Default	Description
Bolt Size	M12	Standard metric sizes from M6 to M20. The tensile stress area (A_s) updates automatically upon selection.
Strength Grade	8.8	ISO 898-1 grades: 4.6 / 5.8 / 8.8 / 10.9 / 12.9. Proof, Re, and Rm values are loaded automatically.
Preload Factor k	0.75	F_pre = k × proof × A_s. Typically 0.6–0.9 depending on tightening method. ~0.75 is typical for torque wrenches.
Preload Loss Factor	0.10	Fraction of preload lost due to relaxation and re-seating after tightening. 10% is a typical value.

ℹ **A data box immediately below the selection fields displays bolt properties in real time.**
Example: M16 / 8.8 → A_s = 157 mm² | proof = 580 MPa · Re = 640 MPa · Rm = 800 MPa

Built-in Bolt Database

Size	d (mm)	A_s (mm²)	Grade	Proof (MPa)	Re (MPa)	Rm (MPa)
M6	6	20.1	4.6	225	240	400
M8	8	36.6	5.8	380	400	500
M10	10	58.0	8.8	580	640	800
M12	12	84.3	10.9	830	900	1000
M14	14	115.0	12.9	970	1080	1200
M16	16	157.0	—	—	—	—
M18	18	192.0	—	—	—	—
M20	20	245.0	—	—	—	—

3.3 Safety Factors

Parameter	Default	Description
SF — Separation	1.5	Safety factor against joint separation. Required total preload = SF_sep × F_pressure.
SF — Stress	1.2	Allowable bolt stress = proof / SF_stress.
Spacing Factor (× d)	3.0	DESIGN mode only. Minimum centre-to-centre spacing between adjacent bolts = factor × bolt diameter. Typical value: 3d.

3.4 Stiffness Model (Toggle)

This section determines the stiffness ratio used to distribute the load between bolt and joint. When the toggle is off, a constant C value is used. When on, the VDI 2230 method is activated.

State	Parameter	Description
Off (default)	Constant C (default: 0.25)	F_bolt = F_pre + C × (F_p / n). Simple and fast.
On (VDI φ)	E_bolt, E_joint, L_eff	φ = kb / (kb + kj). More accurate, based on bolt and flange stiffness.

ℹ **Thermal analysis becomes available only when the VDI φ mode is active. Enabling it is mandatory for thermal calculations.**

VDI Stiffness Parameters:

Parameter	Default	Description
E_bolt (MPa)	210 000	Elastic modulus of the bolt material. 210 000 MPa for steel.
E_joint (MPa)	70 000	Elastic modulus of the flange material. ~70 000 MPa for aluminium, 210 000 MPa for steel.
L_eff — Effective Bolt Length (mm)	50	Effective load-bearing length of the bolt, typically the clamping length.

3.5 Thermal Loading (Toggle)

When bolt and flange materials have different coefficients of thermal expansion, temperature changes alter the bolt preload. This section accounts for that variation.

⚠ **Thermal analysis is only available when the VDI φ stiffness mode is active. Enabling the toggle without VDI φ will not apply any thermal correction.**

Parameter	Default	Description
ΔT (°C)	100	Temperature difference between assembly and operating conditions. Positive = heating.
L_joint (mm)	50	Length of the flange/joint component that participates in thermal expansion.
α_bolt (×10⁻⁶ /K)	12	Coefficient of thermal expansion of the bolt material. Steel: ~12 µm/(m·K).
α_joint (×10⁻⁶ /K)	23	Coefficient of thermal expansion of the flange material. Aluminium: ~23 µm/(m·K).

The calculated thermal load differential (ΔF_th) is added to the effective preload and shown separately in the results panel. A positive value indicates a preload increase; a negative value indicates preload loss.

3.6 Gasket Seating Check (Toggle)

Checks whether the compressive stress on the gasket surface is sufficient for leak tightness. Enabling the toggle reveals a single additional parameter.

Parameter	Default	Description
Min. Gasket Stress (MPa)	20	Minimum contact pressure required for the gasket to seal. Obtained from the gasket manufacturer or the applicable standard.

The tool calculates the gasket contact stress using: p_gasket = (Total Preload − Pressure Force) / A_gasket, and compares it to the specified minimum.

4. Calculation Engine — Formulas

The following table summarises the sequential calculation steps performed in the background.

#	Step	Formula	Description
1	Effective diameter & area	$d_{eff} = (d_{in} + d_{out}) / 2$ $A_{eff} = \pi/4 \times d_{eff}^2$	Area on which pressure force acts
2	Pressure force	$F_p = P \times A_{eff}$	Total pressure load to be carried by the bolts
3	Nominal preload	$F_{pre_raw} = k \times proof \times A_s$	Nominal clamping force (before loss correction)
4	Effective preload	$F_{pre} = F_{pre_raw} \times (1 - loss)$	Preload after relaxation correction
5	Stiffness factor	$\varphi = kb/(kb+kj)$ or constant C	VDI or constant stiffness ratio
6	Thermal correction	$\Delta F_{th} = (\alpha j \cdot Lj - \alpha b \cdot Lb) \times \Delta T / (1/kb+1/kj)$	Applied only when VDI + Thermal are active
7	Load per bolt	$F_{bolt} = F_{pre} + \varphi \times (F_p / n)$	Total force on one bolt under operating conditions
8	Bolt stress	$\sigma = F_{bolt} / A_s$	Governing quantity for the stress check

5. Interpreting Results

5.1 Overall Status Banner

The coloured banner at the top of the right panel summarises the outcome of all checks.

Status	Meaning
✓ **ACCEPTABLE**	All checks passed. The design is adequate.
✗ **NOT ACCEPTABLE**	At least one check failed. Review the details below.

5.2 Metric Cards

Six metric cards appear after the analysis runs. The colour bar on the left edge of each card indicates the category.

Card	Colour	Content
Effective Area	Blue	A_{eff} (mm²) — Calculation area on which the pressure force acts
Pressure Force	Orange	F_p (kN) — Total force exerted by pressure across the entire flange
Effective Preload	Green	F_{pre} (kN) — Per-bolt preload after loss correction
Load per Bolt	Blue	F_{bolt} (kN) — Total force carried by one bolt under operating conditions
Stiffness Factor	Purple	φ or C — Load distribution ratio (mode label is shown)
Bolt Count	Green	n — Value entered in CHECK mode or calculated in DESIGN mode

5.3 Check Results

Each check displays a green OK or red NOK badge in the top-right corner of its block.

[1] **Bolt Stress Check**
Condition: $\sigma = F_{bolt} / A_s \leq proof / SF_{stress}$
A stress utilisation bar shows the ratio: green (< 65%), amber (65–85%), red (> 85%).

[2] **Joint Separation Check**
Condition: $n \times F_{pre} > SF_{sep} \times F_p$
A positive margin (green) confirms the joint remains closed under pressure. A negative margin is shown in red.

[3] **Gasket Seating Check (optional)**
Condition: $p_{gasket} = (n \times F_{pre} - F_p) / A_{gasket} \geq p_{min}$
Visible when the Gasket Seating toggle is on. Gasket area and stress values are listed in the sub-boxes.

[4] **Bolt Spacing Check (DESIGN mode only)**
Condition: Calculated pitch ≥ minimum pitch (= spacing factor × d)
Calculated pitch = π × BCD / n. If the spacing is insufficient, n is incremented and the check is repeated.

5.4 DESIGN Mode — Additional Results

An additional section appears below the check blocks in DESIGN mode:

- Minimum n (raw): Absolute minimum bolt count based on the separation safety factor alone.
- Selected n: Recommended final bolt count that also satisfies the spacing condition.
- Calculated Pitch (mm): Centre-to-centre distance between adjacent bolts. Green = adequate, red = insufficient.

6. Worked Example — Step by Step

ℹ **This example demonstrates the CHECK mode verification of a pressure vessel flange at 0.5 MPa internal pressure.**

Step 1: Select the Mode

Click the CHECK button in the top-right corner (selected by default).

Step 2: Enter Pressure & Geometry

Internal Pressure: 0.5 MPa
Bolt Count: 12
Gasket Inner Diameter: 200 mm
Gasket Outer Diameter: 240 mm

Step 3: Select Bolt Properties

Bolt Size: M16
Strength Grade: 8.8
Preload Factor k: 0.75
Preload Loss Factor: 0.10

The data box updates immediately: M16 / 8.8 → A_s = 157 mm² · proof = 580 MPa · Re = 640 MPa

Step 4: Enter Safety Factors

SF — Separation: 1.5
SF — Stress: 1.2

Step 5: Enable Optional Modules (if required)

For this example, enable the Gasket Seating check:
→ Toggle Gasket Seating on
→ Set Min. Gasket Stress = 25 MPa

Step 6: Run the Analysis

Click the blue 'RUN ANALYSIS' button on the left. Results appear instantly in the right panel.

Step 7: Interpret the Results

Green 'ACCEPTABLE' banner → all checks passed.
Stress Check: review the σ value and utilisation percentage.
Separation Check: confirm that the margin is positive and sufficiently large.
Gasket Seating Check: confirm that p_gasket ≥ 25 MPa.

7. Error Messages

In the event of invalid input, a red warning box appears at the bottom of the left panel.

Error Message	Resolution
Inner diameter must be less than outer diameter.	Verify that the Gasket Inner Diameter is strictly less than the Outer Diameter.
BCD must be entered in DESIGN mode.	Fill in the BCD field when operating in DESIGN mode.
Enter a valid bolt count.	In CHECK mode, the Bolt Count field must contain a positive integer.

8. Limitations & Recommendations

Tool Limitations

- Supported bolt size range: M6–M20.
- All geometries assume cylindrical symmetry.
- The stiffness model follows the simplified VDI 2230 approach; for non-standard flange geometries, FEM analysis is recommended.
- Thermal analysis is not available in constant-C mode; VDI φ mode is required.
- Gasket seating stress is calculated based on an average contact stress assumption; actual pressure distribution may differ.

Design Recommendations

- Always activate the Thermal Loading module for elevated-temperature applications.
- Compare different bolt size and strength grade combinations to identify the optimum selection.
- Cross-validate the bolt count n obtained in DESIGN mode by re-checking it in CHECK mode.
- Use dimensions from standard flange series (DN, ANSI) as inputs for BCD and gasket diameters.
- Before reducing safety factors, verify consistency with the applicable standard (VDI 2230 / EN 1591).

9. Quick Reference

Toggle Summary

Toggle	Dependency	What It Adds	Default
VDI φ Stiffness	Independent	E_bolt, E_joint, L_eff → φ formula	Off
Thermal Loading	VDI φ must be active	ΔT, L_joint, α_bolt, α_joint → ΔF_th	Off
Gasket Seating	Independent	p_min → gasket stress check [3]	Off

Default Parameter Summary

Parameter	Default	Context
Internal Pressure	0.25 MPa	All modes
Bolt Count	8	CHECK mode
Bolt Size / Grade	M12 / 8.8	All modes
Preload Factor k	0.75	All modes
Preload Loss Factor	0.10	All modes
Stiffness Constant C	0.25	When VDI is off
SF — Separation	1.5	All modes
SF — Stress	1.2	All modes
Spacing Factor	3.0 × d	DESIGN mode
Min. Gasket Stress	20 MPa	When Gasket Seating is on